Ground-Based Distributed GNSS Spoofing of UAV Swarms: Formation Dilation and Its Detection by Relative-Distance Tests

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[TODO: affiliazione, email]

***Abstract*—Multi-receiver spoofing detectors for UAV swarms assume that the spoofer has a single transmission point, so that an attack collapses the inter-UAV distances computed from GNSS. We study a distributed ground spoofer that breaks this assumption: 24 fixed transmitters on a 10 km circle around the operating area, each replaying one satellite with the authentic navigation message, simulated transmitter by transmitter with three-dimensional propagation and a ground-to-UAV channel model. The attack displaces a 25-UAV swarm by about 4 km, but it does not preserve the formation. Because ground transmitters cannot match the satellite elevations, the computed formation is *dilated*, by a median of 22 % of the baseline across missions, and a first-order least-squares model reproduces this dilation epoch by epoch. Since a dilation increases the inter-UAV distances, the collapse tests of Jansen et al. never fire, and a worst-pair GNSS-vs-UWB check detects at most 26 % of the epochs at a per-epoch false-alarm rate of 10^{-2} over formation spacings of 5–20 m. A two-sided test on the median ratio between GNSS-derived and UWB-measured distances detects the dilation in 38 % of the epochs at 5 m and in 98 % at 20 m. Raising the transmitters first shrinks the dilation and then reverses it, with a zero crossing at about 7 km of altitude for a 10 km array. Only transmitters placed on the satellites' lines of sight, at several kilometres of slant range, are detected in fewer than a quarter of the epochs.**

***Index Terms*—GNSS spoofing, UAV swarms, spoofing detection, multi-receiver, distributed attack, relative distances, formation flight.**

I. Introduction

The open civilian GNSS signals carry no signal-level authentication. Their structure and navigation message are public, and a transmitter that reproduces them with enough power can make a receiver compute any position and time of the attacker's choosing. Navigation-message authentication, now operational on Galileo [1], does not prevent this: an attacker that replays the authentic message and forges only the signal delays, as in this paper, passes the check. Since the first portable civilian spoofer was demonstrated [2], software-defined radios and open-source signal generators have made the attack accessible for a few hundred dollars, and incidents affecting ships, aircraft and unmanned systems are reported routinely [3]. Unmanned aerial vehicles (UAVs) are a natural target: they navigate on GNSS by default, and their capture by spoofing has been demonstrated in the field [4]. A swarm flying in formation, for surveying, inspection, delivery or defense, multiplies both the value of the target and the consequences of a successful hijack.

Among the countermeasures proposed in the last fifteen years [3], [5], *multi-receiver* detection stands out for its simplicity and for the strength of its guarantee. Its principle, established by Tippenhauer et al. [6], is geometric. A spoofer with a single antenna delivers the same signals to every receiver in its coverage area, so all the victims compute the *same* position fix. A group of receivers that knows its own formation can therefore detect the attack by checking that the mutual distances implied by the GNSS fixes have not collapsed. Jansen et al. [7] turned the check into a calibrated statistical test, showing that four receivers a few metres apart reach a detection probability of 0.99 at a false-alarm rate of 10^{-5}. Swaszek and Hartnett [8] derived Neyman–Pearson optimal tests for the same setting. For UAV swarms, the same idea has been proposed with the formation measured on-line by inter-drone ultra-wideband (UWB) ranging [9], or verified as a localization feasibility problem [10]. The guarantee rests on one assumption: the attacker has a single transmission point.

This paper asks what happens when this assumption is dropped with a *physical* distributed spoofer rather than an idealized attacker that imposes a common offset on every victim. The spoofer consists of several fixed ground transmitters surrounding the swarm's operating area, each impersonating one satellite, subject to the real propagation geometry and to a ground-to-UAV channel. Formation-preserving spoofing has been demonstrated in hardware when the transmitters are placed along the satellite directions at altitude [11]. Ground transmitters, however, cannot reproduce the satellites' elevation, and neither the resulting deformation of the computed formation nor the relative-distance tests able to observe it have been quantified.

We find that a ground array of two dozen fixed radios does not collapse the formation but dilates it. The collapse tests of [7] therefore never fire, and the worst-pair GNSS-vs-UWB check of [9] misses most epochs at standard GNSS noise. With the same UWB measurements as [9], a different statistic, a two-sided test on the median distance ratio, recovers much of the detection. We study the inter-UAV spacing of 5 to 20 m typical of outdoor GNSS-based formation flight (Section V-A). Our contributions are:

- a transmitter-by-transmitter simulation of a distributed ground spoofer attacking a moving UAV swarm, with per-epoch routing of satellites to transmitters, three-

dimensional propagation of the replayed signals, and the elevation-dependent ground-to-UAV path-loss model of [12] (Section III-B);
- the finding that a ground array *dilates* the computed formation, by a median of 22 % of the baseline across missions, because of the mismatch between satellite and transmitter elevations, together with a first-order least-squares model that predicts the dilation epoch by epoch (Sections III-C and V-C);
- a Monte Carlo evaluation, with mission-level bootstrap confidence intervals, of five relative-distance detectors: the three collapse variants of [7], the worst-pair GNSS-vs-ranging check [9], and a two-sided test on the median distance ratio. The dilation escapes the first four and is detected by the fifth in most epochs from 8 m and reliably from 15 m (Section V-D);
- a transmitter-altitude sweep and a static line-of-sight (WASP-like) placement experiment that measure what it costs the attacker, in altitude, to remove the dilation: the signed dilation crosses zero at a transmitter elevation of about 36° (7 km of altitude for a 10 km array), and a static constellation on the satellites' lines of sight at ≥ 5 km slant range is detected in fewer than a quarter of the epochs (Section V-E).

The simulator and the detection code are released with the paper.

II. Related Work

A. Multi-receiver detection in the position domain

Tippenhauer et al. [6] formalized the GPS group-spoofing problem and proved two complementary results. A single-antenna spoofer places all victims at the same location (Result 1), and a formation-preserving placement of one transmitter per impersonated satellite always exists, along the satellite directions (Result 7). Their countermeasure checks that the computed positions preserve the physical formation, without a statistical model. Jansen et al. [7] supplied the model: the pairwise distances between receivers are Rician under authentic conditions and Rayleigh under a single-transmitter attack, with spatially correlated position noise ($\sigma \approx 0.5$–3 m, $\rho \approx 0.4$–0.6 measured). Their test $f(\{d_{ij}\}) < \lambda$ is evaluated in minimal, maximal, majority and normalized-majority variants, the last being the best. Their prototype with four receivers on a 5 m square ran for three hours without errors, and their own attempt at a distance-preserving attack with four USRPs succeeded on one receiver and failed on two. Swaszek and Hartnett [8] derived Neyman–Pearson optimal tests for COTS receivers at known relative positions, and later fused GNSS fixes with inter-vehicle ranges for platoons under the assumption that only one vehicle is spoofed [13]. Crowd-GPS-Sec [14] applies the collapse principle to crowdsourced air-traffic receivers, combined with an absolute time-difference-of-arrival reference.

B. Measurement-domain and swarm variants

In the measurement domain, Stenberg et al. [15] test double differences of pseudorange and carrier phase between receivers with a χ^2 GLRT, and find a minimum receiver separation of about 5 m with standard positioning. Kalantari and Larsson [16] derive a GLRT against multiple spoofers for receivers on a rigid body with unknown pose, and show that more receivers make the attack easier to detect. Borio and Gioia [17] detect the single common direction of arrival with a two-receiver carrier-phase sum-of-squares statistic. For UAV swarms, Mykytyn et al. [9] compare GNSS-derived inter-drone distances with UWB two-way ranging and raise an alarm when they differ by more than about 5 m; they consider a coordinated per-drone attack impractical. Bi et al. [10] cast detection as a semidefinite localization-feasibility problem that is robust to colluding members. GNSS/INS fusion defenses, such as SemperFi [18] and tightly coupled INS innovation monitors [19], [20], compare the GNSS track with inertial dead reckoning and respond to the time variation of the induced error.

C. Coordinated and distributed spoofing

Attack-side work has progressively approached the formation-preserving placement of [6]. We order it by *where the transmitters sit*, because this decides the residual geometry.

Single transmitter. Beyond Result 1 of [6], Chen et al. [21] analyse a single spoofer against several receivers and show that preserving the formation *exactly* is possible only for a small group (at most about nine receivers); beyond that, the formation is deformed and a relative-distance check detects it. A follow-up [22] develops approximate constructions that spoof more receivers at the price of a bounded error. A single transmitter therefore either collapses the swarm or scales poorly, which motivates distributing the transmitters.

Ground arrays. Ceccato et al. [23] designed a multi-antenna ground spoofer with Wiener precoding that displaces a swarm over a target plane while minimising a mean-squared position error. It is a signal-processing model, not simulated on a flying swarm and not confronted with a detector; GNSS-WASP [11] lists the wrong angle of arrival of its ground-based signals as its main limitation. Cheng et al. [24] realised a distributed multi-antenna spoofer with off-the-shelf radios, showed that it requires sub-50 ns alignment between transmitters, and steered receivers in an anechoic chamber with an average *position* error of about 30 m. Neither work studies what a ground array does to the *formation* of a swarm, nor whether a relative-distance detector sees it.

Airborne constellations. GNSS-WASP [11] demonstrated in hardware a constellation of at least four synchronised transmitters placed *in line with the satellites at altitude*. It spoofs receivers over a square-kilometre area independently of their number, and preserves their relative distances with an average error of about 1 m in its reference configuration, bypassing multi-receiver [7] and sensor-fusion [18] countermeasures. Its authors note that the residual bias of a real constellation

decreases with the transmitter altitude but cannot be eliminated short of placing the transmitters in orbit. They also evaluate an attack-aware countermeasure that flags an outlier among the pairwise distance residuals, after a moving average or a CUSUM over time. Temporal accumulation suits a residual whose sign varies from pair to pair. The residual of a *ground* array instead has a coherent sign (a dilation), which allows aggregation *over the pairs* (Section IV-A).

Idealised limit. Park and Yoo [25] state the translation invariance of relative-only detectors as a formal proposition, rooted in the gauge freedom of anchor-free network localization [26], and evaluate it by simulation with an idealised common offset rather than a physical transmitter layout. Surveys of UAV spoofing defenses [5], [27] note that detectors relying on neighbours' positions fail when the neighbours are inside the spoofed area, but do not quantify the effect for a physical ground array.

D. Positioning of this work

The ground-array case is the one left uncharacterised: it has been modelled [23] and built [24], but its effect on a formation and on relative-distance detectors has not been measured. We study a ground array with one transmitter per satellite, the canonical placement of [6, Result 7], against a flying swarm, and we quantify how far it is from the ideal formation-preserving attacker of [11], [25].

III. System and Distributed Spoofing Attack Model

A. UAV Swarm and Navigation Model

We consider a swarm of $m = 25$ UAVs flying at altitude h in a square 5×5 horizontal grid with spacing g (the *formation spacing*) along a way-point trajectory (Table I). The swarm is served by a GNSS constellation of satellites in medium Earth orbit. For UAV u at position $\mathbf{p}_u(t)$ and satellite s at position $\mathbf{s}_s(t)$, visible when its elevation is positive, the authentic propagation delay is $\tau_{u,s}(t) = \|\mathbf{s}_s(t) - \mathbf{p}_u(t)\|/c$. Each receiver has a constant clock bias δt_u and independent additive white Gaussian pseudorange noise of standard deviation $\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, representative of standard (non-differential) positioning.

Each UAV computes its position fix $\hat{\mathbf{p}}_u$ independently from the measured delays $\tilde{\tau}_{u,s}$ (authentic or spoofed) with the standard iterated linearized least-squares (Gauss–Newton) solution over the four unknowns $(x, y, z, \delta t)$; without attack and without noise it recovers the true grid positions. In addition to GNSS, the UAVs measure their mutual distances with an independent on-board ranging sensor (UWB two-way ranging, as in [9]):

$$d_{ij}^{\mathrm{meas}}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| + \varepsilon_{ij}(t), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_{\mathrm{rng}}^2), \tag{1}$$

with $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$ and no ranging bias.

The UAVs follow their planned trajectory regardless of their GNSS fix; the closed-loop reaction of guidance and formation control to the spoofed fixes is not modelled (Section VI).

B. Distributed Spoofing Attack

The attacker moves the swarm's perceived position onto a fixed decoy point, at about 4 km from the true trajectory, while preserving the perceived formation as far as its geometry allows. The attack is simulated transmitter by transmitter, with no per-victim signal shaping. As in [11], the spoofer replays the authentic navigation message, so the receivers use the true satellite ephemerides and only the code phases are forged.

Transmitter geometry and per-epoch routing. $N_{\mathrm{sp}} = 24$ fixed ground transmitters are placed on a circle of radius $R_{\mathrm{sp}} = 10\,\mathrm{km}$ that encloses the swarm's operating area (Fig. 2), each with an EIRP of 2 dBm (1 mW, 1 dBi antenna gain). Following the geometric argument of [6, Result 7], each visible satellite s is routed, at every epoch, to the transmitter whose azimuth ϕ_k, as seen from the *current* true swarm centre $\mathbf{p}^{\mathrm{ctr}}(t)$, is closest to the satellite azimuth ϕ_s:

$$\mathrm{sp}(s,t) = \arg\min_k \left| \angle_{2\pi}\big(\phi_s(t) - \phi_k(t)\big) \right|, \tag{2}$$

where $\angle_{2\pi}(\cdot)$ wraps an angle to $(-\pi, \pi]$. With 24 transmitters the azimuth mismatch is at most 7.5°. The transmitters never move; only the routing is updated, which requires the attacker to track the true position of the swarm centre, the same knowledge it needs to calibrate the replayed delays.

When a satellite is re-routed, the replayed delay changes abruptly for every UAV except the centre, by the difference between the two propagation residuals of (4). The simulator applies no smoothing, so this is the worst case for the attacker. Re-routing is rare: at $g = 10\,\mathrm{m}$ it affects 2.2 % of the satellite-epochs (0.15 satellites per epoch on average), because consecutive transmitters are 15° apart in azimuth while the satellites move slowly. Its effect on the computed formation is visible but occasional: the per-epoch dilation changes by 0.5 percentage points between consecutive epochs in the median, with a largest jump of 12.8 points over a mission. A real attacker could hand a satellite over gradually, which would remove these transients.

Ground-to-UAV channel. The spoofing signals travel over a ground-to-air link, modelled with the suburban cellular-to-UAV path loss of Al-Hourani and Gomez [12], which we also adopted in earlier work [28]. For a transmitter at horizontal distance d (in metres) from the UAV, with θ the elevation angle of the UAV as seen from the transmitter (in degrees),

$$\mathrm{PL}(d,\theta) = 10\,\alpha \log_{10}(d) + A\,(\theta - \theta_0)\, e^{-\frac{\theta-\theta_0}{B}} + \eta_0 \quad [\mathrm{dB}], \tag{3}$$

with $\alpha = 3.04$, $A = -23.29$, $B = 4.14$, $\theta_0 = -3.61^\circ$ and $\eta_0 = 20.70\,\mathrm{dB}$; shadowing is disabled for repeatability. The excess-loss term is fitted in [12] for $\theta \in [-2^\circ, 10^\circ]$ and saturated at 10° beyond. In our deployment the largest angle met in any run is 6.0°, so the model is always used within its validity range. The model was fitted on cellular bands; we use it for the GNSS L1 band as a representative ground-to-air loss.

Signal capture and replayed delay. For each UAV u and visible satellite s, the receiver locks onto the transmitter whenever the received spoofing power, computed with (3),

exceeds the authentic one. With the chosen EIRP the spoofer margin over the satellite signal is between 13 and 29 dB on every link (median 19.5 dB), so capture is complete and homogeneous across the swarm: no UAV mixes authentic and spoofed satellites.

Transmitter $k = \mathrm{sp}(s,t)$ replays the signal of satellite s with its code phase calibrated so that, at the true swarm centre, the measured delay equals the decoy delay τ_s^{dec}, i.e. the delay that satellite s would exhibit at the decoy position. A UAV at a different position receives the same transmission over a different path and measures

$$\tilde{\tau}_{u,s} \;=\; \tau_s^{\mathrm{dec}} \;-\; \delta t_u \;+\; \frac{\|\mathbf{q}_k - \mathbf{p}_u\| - \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|}{c}, \qquad (4)$$

where $\mathbf{q}_k$ is the transmitter position and both norms are true three-dimensional slant distances. The last term is the propagation residual: it vanishes for the centre UAV, which computes exactly the decoy position, and is nonzero for the other UAVs. It does not depend on the transmitted power, so once capture is complete the residual is independent of it.

C. Formation Deformation under Distributed Spoofing

Write $\mathbf{p}_u = \mathbf{p}^{\mathrm{ctr}} + \boldsymbol{\delta}_u$ with $\|\boldsymbol{\delta}_u\| \ll \|\mathbf{q}_k - \mathbf{p}^{\mathrm{ctr}}\|$, and let $\hat{\mathbf{v}}_k$ be the unit vector from the centre to transmitter k and $\hat{\mathbf{e}}_s$ the unit line of sight from the centre to satellite s. To first order, the residual delivered by (4) is $-(\hat{\mathbf{v}}_k \cdot \boldsymbol{\delta}_u)/c$. The delay that would make UAV u compute the decoy position *rigidly displaced* by $\boldsymbol{\delta}_u$ is instead $-(\hat{\mathbf{e}}_s \cdot \boldsymbol{\delta}_u)/c$. The pseudorange error with respect to the rigid attack, in metres, is therefore

$$\epsilon_{u,s} \;=\; (\hat{\mathbf{e}}_s - \hat{\mathbf{v}}_k) \cdot \boldsymbol{\delta}_u \;\approx\; -(\hat{\mathbf{a}}_s \cdot \boldsymbol{\delta}_u)\left(\cos\theta_k - \cos \mathrm{el}_s\right), \quad (5)$$

where $\hat{\mathbf{a}}_s$ is the horizontal unit vector in the satellite azimuth, θ_k the elevation of transmitter k as seen from the swarm, and el_s the satellite elevation. The approximation holds for a horizontal swarm and neglects the azimuth mismatch of the routing (2); the exact first-order expression on the left retains it.

For a ground array far from the swarm, $|\theta_k|$ is a few degrees, so $\cos\theta_k \approx 1 > \cos \mathrm{el}_s$ and the coefficient in (5) is negative for every satellite (Fig. 1). A UAV displaced *towards* the azimuth of satellite s measures a pseudorange *shorter* than the rigid one, and a UAV displaced away from it a longer one. The least-squares fix interprets shorter ranges to the satellites ahead and longer ranges to those behind as a position farther along $\boldsymbol{\delta}_u$. Every UAV is thus pushed outward along its own displacement from the centre, and the computed formation is *dilated*. The coefficient vanishes only when $|\cos\theta_k| = \cos \mathrm{el}_s$, i.e. for satellites near the horizon or when $|\theta_k|$ approaches el_s. Since the cosine is even, this happens either for transmitters raised in altitude (Section V-E) or, symmetrically, for transmitters seen at a comparable *depression* angle, i.e. close to the swarm and well below it. We verified the second branch by placing the array 1 km below the swarm: at horizontal ranges of 5, 2 and 1 km the median transmitter elevation is $-14.7°$, $-34.9°$ and $-58.1°$, and the signed dilation is $+22.0\,\%$, $+5.4\,\%$ and $-22.7\,\%$. It crosses zero near $|\theta_k| = 35°$, where $\cos\theta_k$ matches the mean $\cos \mathrm{el}_s$ of the constellation, as (5) predicts. A low array very close to the swarm is therefore an alternative to altitude, at the cost of a coverage area of the same size as the stand-off distance.

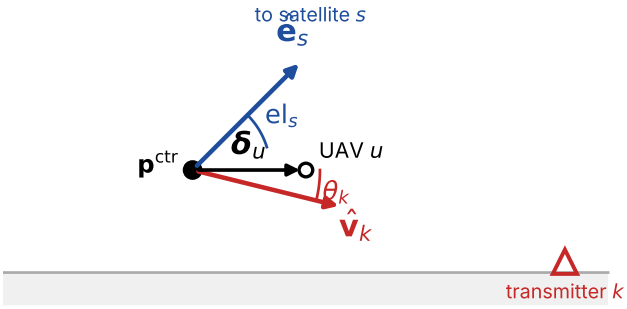


Fig. 1. Geometry of the residual (5), seen from the swarm centre. The replayed signal of satellite s arrives from the direction $\hat{\mathbf{v}}_k$ of its ground transmitter instead of the satellite direction $\hat{\mathbf{e}}_s$. For a UAV displaced by $\boldsymbol{\delta}_u$ the two directions project differently onto $\boldsymbol{\delta}_u$, and since $\cos\theta_k > \cos \mathrm{el}_s$ the pseudorange error is negative towards the satellite azimuth: the fix is pushed outward and the formation is dilated.

Predicted dilation. Let $\mathbf{G}$ be the geometry matrix of the four-unknown fix, with rows $[-\hat{\mathbf{e}}_s^{\mathsf{T}}, 1]$, and $\mathbf{B}$ the matrix with rows $(\hat{\mathbf{e}}_s - \hat{\mathbf{v}}_{\mathrm{sp}(s,t)})^{\mathsf{T}}$, so that the error vector of (5) is $\boldsymbol{\epsilon}_u = \mathbf{B}\boldsymbol{\delta}_u$. The linearized least-squares response to this error gives the computed position

$$\hat{\mathbf{p}}_u = \mathbf{p}^{\mathrm{dec}} + (\mathbf{I} + \mathbf{M})\,\boldsymbol{\delta}_u, \qquad \mathbf{M} = \left[(\mathbf{G}^{\mathsf{T}}\mathbf{G})^{-1}\mathbf{G}^{\mathsf{T}}\mathbf{B}\right]_{1:3}, \quad (6)$$

where $[\cdot]_{1:3}$ keeps the three position rows and discards the clock row. The predicted computed distance of each pair is $\|(\mathbf{I} + \mathbf{M})(\boldsymbol{\delta}_i - \boldsymbol{\delta}_j)\|$, including the vertical component of the error. We refer to (6) as the *first-order model*. A cruder estimate follows by matching (5), for a single satellite, with the position response $-\hat{\mathbf{e}}_s \cdot \Delta\mathbf{p}$ under the assumption $\Delta\mathbf{p} \parallel \boldsymbol{\delta}_u$, which gives $\Delta\mathbf{p} = (1/\cos \mathrm{el}_s - 1)\,\boldsymbol{\delta}_u$. Averaged over the constellation, this yields the dilation scale

$$\kappa_0 \;=\; \frac{1}{\langle \cos \mathrm{el}_s \rangle} - 1, \qquad (7)$$

which is 0.3–0.4 for typical elevation distributions. In our runs κ_0 overestimates the dilation, because the clock and vertical unknowns of the fix absorb part of the error (Section V-C); it is useful only as an order of magnitude. Both (6) and (7) are independent of the transmitter power and of the formation spacing g: the dilation is a fixed fraction of the baseline.

IV. Relative-Distance Spoofing Detection

A. Detection Principle

A defender compares the inter-UAV distances implied by the GNSS fixes with the independent UWB ranges through

the normalized ratios

$$r_{ij}(t) = \frac{\hat{d}_{ij}(t)}{d_{ij}^{\mathrm{meas}}(t)}, \qquad \hat{d}_{ij} = \|\hat{\mathbf{p}}_i - \hat{\mathbf{p}}_j\|, \tag{8}$$

computed over all $\binom{m}{2} = 300$ pairs at every epoch, with d_{ij}^{meas} the UWB measurement (1). Under authentic conditions $r_{ij} \approx 1$.

The *sign* of the distortion distinguishes the two attack regimes. A single-transmitter spoofer collapses the fixes onto one point [6, Result 1], so the classical detectors threshold a statistic that *decreases* when the pairwise distances shrink. A ground array with one transmitter per satellite dilates the formation (Section III-C), so the same statistic *increases*, into the tail that a one-sided collapse test does not examine. Because the dilation has the same sign on all pairs, it can be detected by aggregating over the pairs at a single epoch, with no need for temporal accumulation.

B. Detection Statistics

Collapse tests (Jansen et al.). The detector of [7] declares spoofing when the pairwise distances *shrink*:

$$\text{alarm if } f\big(\{r_{ij}\}\big) < \lambda, \tag{9}$$

with the three variants of its Appendix A:

$$f_{\min} = \min_{i<j} r_{ij}, \quad f_{\max} = \max_{i<j} r_{ij}, \quad f_{\mathrm{maj}} = \operatorname*{med}_{i<j} r_{ij}. \tag{10}$$

$f_{\min}$ alarms if any pair has shrunk, $f_{\max}$ only if all pairs have, and f_{maj} if at least half of the pairs have; the latter is equivalent to the normalized majority vote of [7], since at least half of the normalized distances are below λ if and only if their median is. Jansen et al. set λ from a Rician (H_0) / Rayleigh (H_1) model of the pairwise distance with spatially correlated position noise; we calibrate λ empirically (Section IV-C).

Two-sided tests. These fire on a deviation of the distance ratios from one in either direction. The *worst-pair* test

$$T_{\mathrm{wp}} = \max_{i<j} \big|\, r_{ij} - 1 \,\big|, \qquad \text{alarm if } T_{\mathrm{wp}} > \lambda, \tag{11}$$

is the ratio form of the GNSS-vs-UWB check of [9], which alarms if $|d_{\mathrm{GPS}} - d_{\mathrm{UWB}}| > d_{\mathrm{thr}}$ on any pair, and the per-epoch core of the outlier test evaluated in [11]. Because it is governed by the noisiest pair, it is weak against a bias shared by all pairs. Against such a bias an *aggregate* statistic is more powerful, such as the χ^2 sum over double differences of [15] or simply the median. We therefore also evaluate the *two-sided majority* test

$$T_{\mathrm{maj}} = \Big|\operatorname*{med}_{i<j} r_{ij} - 1\Big|, \qquad \text{alarm if } T_{\mathrm{maj}} > \lambda, \tag{12}$$

the two-sided counterpart of f_{maj}. We centre both two-sided statistics on one for comparability with [9]; the consequence of the noise bias of the median ratio is discussed in Section V-E. Finally, since the dilation is persistent, both two-sided statistics are also evaluated after a moving average over 12 epochs (1 min) within each mission.

C. Threshold Selection and Metrics

Following the Neyman–Pearson approach of [7], [29], thresholds are calibrated on the H_0 (no-spoofing) scenario simulated with the same noise levels: λ is the empirical P_{FA}-quantile of the statistic over all H_0 epochs (lower quantile for the collapse tests, upper for the two-sided tests), with a per-epoch $P_{\mathrm{FA}} = 10^{-2}$. This rate is higher than the 10^{-5} targeted in [7], because lower rates cannot be estimated reliably from the 4800 H_0 epochs available per spacing.

We report (i) the detection probability P_{D}, the fraction of attacked epochs that raise an alarm, and (ii) the receiver operating characteristic (ROC) curve, the locus of $(P_{\mathrm{FA}}, P_{\mathrm{D}})$ pairs as λ is swept, with its area (AUC) [29]. The AUC is the probability that the statistic of a random attacked epoch falls on the alarm side of that of a random H_0 epoch. An AUC of 0.5 is chance; an AUC near 1 means that the two distributions are separated on the side the test examines; an AUC near 0 means that they are separated on the *opposite* side, i.e. the attack is observable, but only in the tail that the test does not examine.

V. Results

A. Simulation Setup

Table I summarizes the parameters. We sweep the formation spacing over $g \in \{5, 8, 10, 12, 15, 20\}$ m, the range of outdoor formation flight. Experimental leader–follower formations of quadcopters have been flown at 5 m spacing with separation errors of 0.4 m [30]; a design method that derives the minimum safe spacing from the combined navigation and control error yields spacings of a few metres for GNSS-based navigation [31]; the multi-receiver prototype of [7] uses a 5 m square, and the swarm detector of [9] is designed around a 5 m GPS error budget. Spacings below a few metres, as in drone light shows, require centimetre-level RTK positioning and are outside the standard-GNSS regime considered here; wider spacings are typical of coverage and surveillance swarms rather than of formation flight.

For each spacing we simulate $N_{\mathrm{MC}} = 20$ independent missions (random start time, clock biases and noise), each of 240 epochs, with and without the attack. The simulator and the detection code are available in the companion repository.[1] Because the epochs of one mission share clocks, trajectory and satellite geometry, the Monte Carlo unit is the *mission*: 95 % confidence intervals on P_{D} are obtained by a paired bootstrap over missions (500 resamples; the H_0 and attacked runs of a mission share the random seed, and the threshold is re-calibrated in every resample).

Besides the detection metrics, we report two geometric quantities. The *distortion*

$$E(t) = \operatorname*{med}_{i<j} \frac{|\hat{d}_{ij}(t) - d_{ij}(t)|}{d_{ij}(t)} \tag{13}$$

[1] https://github.com/silviasch01/GPS_spoofing_distribuito

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
Swarm size m	25 (5×5 grid)
Formation spacing g	5, 8, 10, 12, 15, 20 m
Flight altitude h	**[TODO: valore]**
UAV speed	**[TODO: valore]**
Mission duration / epoch	1200 s / $T_s = 5$ s
Satellites	24 (6 planes, 55°, 20 200 km)
Elevation mask	0°
Pseudorange noise σ_{pr}	3 m (white, independent)
Ranging noise σ_{rng}	0.1 m (unbiased)
Ground transmitters N_{sp}	24 on circle, $R_{\mathrm{sp}} = 10$ km
Transmitter EIRP	2 dBm (1 mW, 1 dBi)
Ground-to-UAV channel	Eq. (3), [12]
Spoofer power margin (all links)	13.2–29.0 dB
Largest transmitter elevation angle	6.0°
Decoy offset (mission mean)	4.0 km
Target false-alarm rate P_{FA}	10^{-2} per epoch
Monte Carlo missions per spacing	20

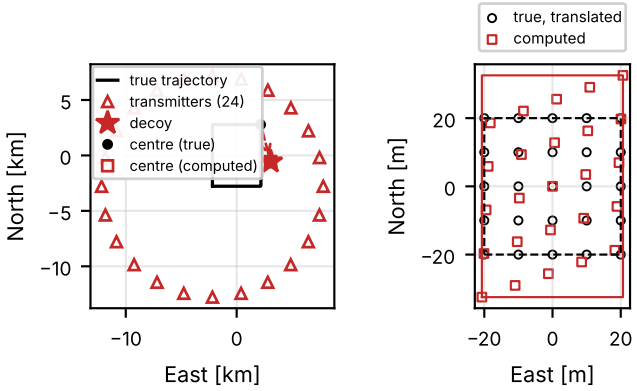


Fig. 2. Attack scenario, one epoch at $g = 10$ m, in a local ENU frame. Left: the swarm flies its mission (black trajectory) inside a 10 km circle of 24 fixed ground transmitters, each replaying one satellite; the computed centre is displaced onto the decoy. Right: the computed grid (red squares) against the true grid translated by the same displacement (black circles), from a noiseless run. The formation is dilated by 18 % at this epoch, not preserved.

is the unsigned evidence available to any relative-distance detector; it includes the noise. The *signed dilation*

$$D = \operatorname*{med}_{t} \operatorname*{med}_{i<j} \frac{\hat{d}_{ij}(t) - d_{ij}(t)}{d_{ij}(t)} \tag{14}$$

is measured on noiseless runs of a mission. It is positive for a dilation and negative for a contraction, and it isolates the effect of the attack geometry.

B. Formation Distortion under Attack

The attack succeeds: every UAV is captured on all its visible satellites, and the computed position of the centre UAV coincides with the decoy at every epoch. Since the decoy is fixed and the swarm moves, the displacement varies along the mission, from about 1 to 6 km (4.0 km on average; Fig. 3, top). The bottom panel shows the price paid by a ground array: the distortion E stays above the no-attack noise floor throughout the mission.

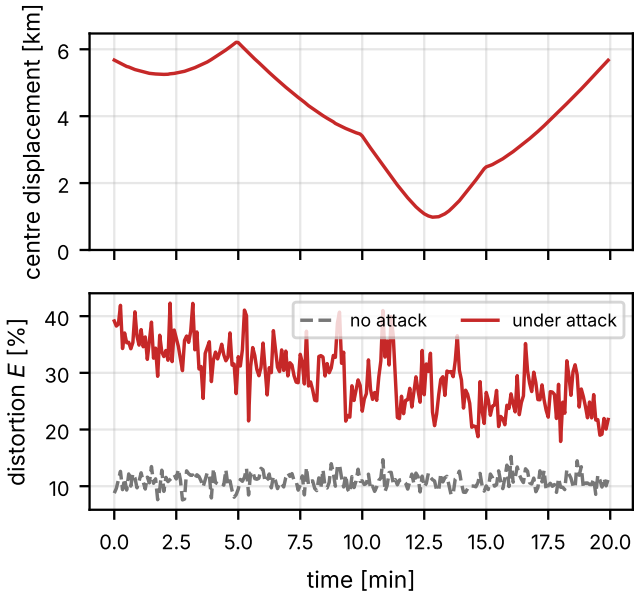


Fig. 3. One mission at $g = 10$ m ($\sigma_{\mathrm{pr}} = 3$ m). Top: distance between the computed and the true position of the centre UAV; it varies because the decoy is fixed while the swarm moves. Bottom: distortion E (13) with and without the attack.

The dilation D has a median of +22 % over the 20 missions at $g = 10$ m (noiseless), but varies widely between missions, from 6.6 to 84 % (10th–90th percentiles), depending on the elevations of the satellites in view. Figure 4 shows the distortion E over the formation spacings. Under H_0, E is pure GNSS noise and decays as $1/g$, from 22 % at $g = 5$ m to 4.6 % at 20 m. Under attack, E decreases from 38 % at 5 m to 23 % at 20 m, approaching the scale-free dilation as the noise contribution fades. The distortion is a dilation, not a collapse: the median distance ratio $\operatorname{med} r_{ij}$ is 1.23 under attack at $g = 20$ m against 1.01 without attack (1.37 against 1.16 at 5 m, where noise biases both upward), and it exceeds one in every attacked epoch.

C. Validation of the First-Order Model

Figure 5 compares the first-order model (6) with the full nonlinear simulation on a noiseless attacked mission at $g = 20$ m. The model predicts a median distortion of 24.4 % against 24.5 % measured, with correlation 1.00 and a largest deviation of 0.4 percentage points, so the linearization holds at these baselines. The comparison of three predictions shows where the dilation comes from. The crude scale (7) gives 34.5 %, the horizontal part of the model 18.8 %, and the full model 24.4 %. The clock and vertical unknowns of the fix absorb part of the pseudorange error (hence 34.5 → 24.4), while the vertical component of the position error, an altitude tilt proportional to $\boldsymbol{\delta}_u$, contributes about a quarter of the three-dimensional distortion (hence 18.8 → 24.4). The dependence of $\mathbf{M}$ on the satellites in view also explains the spread of the dilation across missions in Fig. 4.

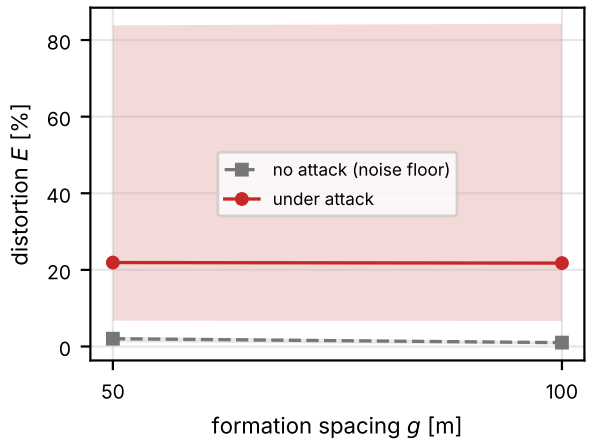


Fig. 4. Distortion E (13) vs. formation spacing (median over 20 missions; band: 10th–90th percentile across missions).

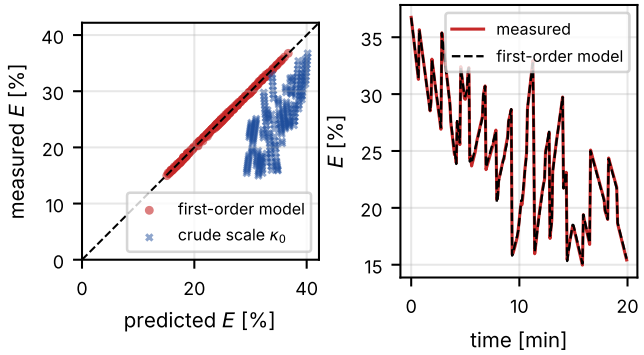


Fig. 5. First-order model vs. simulation on a noiseless attacked mission ($g = 20$ m, 240 epochs). Left: measured vs. predicted distortion E from the first-order model (6) (dots) and from the crude scale κ_0 (7) (crosses); dashed: bisector. Right: measured and predicted E along the mission.

D. Detection Performance

Table II and Fig. 6 report P_{D} at $P_{\mathrm{FA}} = 10^{-2}$ with 95 % confidence intervals, Table III the ROC areas, and Fig. 7 the ROC curves at $g = 10$ m.

Collapse tests. As expected from the sign of the distortion, the three Jansen variants never detect the dilation: $P_{\mathrm{D}} \leq 0.011$ (95 % upper bound) at every spacing, at or below the false-alarm rate. Their AUC is *below* chance, down to 0.001 for the majority variant at 20 m. This does not mean that the attack is invisible: for the majority variant, the attacked statistic is larger than the authentic one in almost every epoch from 10 m upward, i.e. the two are separable, but on the side that a collapse test does not examine. The majority variant, the one recommended in [7], never fires, and no choice of threshold changes this.

Worst-pair test. The worst-pair test is masked by noise: P_{D} rises only from 0.03 at 5 m to 0.26 at 20 m (AUC 0.57–0.74), with wide confidence intervals that reflect the variability of the dilation across missions. The statistic (11) is governed by

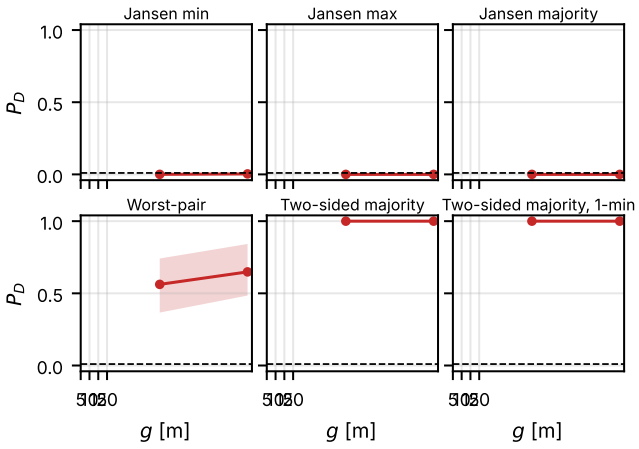


Fig. 6. Detection probability vs. formation spacing at $P_{\mathrm{FA}} = 10^{-2}$ per epoch (20 missions; band: 95 % bootstrap CI; dashed: P_{FA}). The collapse detectors never leave the P_{FA} line; the two-sided majority test detects the dilation reliably from 15 m, and from 12 m with one-minute integration.

the noisiest of 300 pairs, and at $\sigma_{\mathrm{pr}} = 3$ m the H_0 tail of that pair is wider than the dilation of a typical pair over the whole spacing range. A one-minute moving average raises P_{D} only to 0.12–0.38 (Table IV).

Two-sided majority test. The aggregate statistic (12) averages the noise over all pairs and retains the common bias, which is the signature of a ground array. It detects the dilation in 0.38 of the epochs at 5 m, 0.57–0.71 at 8–12 m, and reliably from 15 m (0.85 at 15 m, 0.98 at 20 m), with AUC from 0.83 to 0.999. At 10 m its ROC (Fig. 7) mirrors that of the collapse majority test: the same statistic, tested in the other tail. With one-minute integration it reaches $P_{\mathrm{D}} = 0.52$ at 5 m, 0.80 at 10 m, 0.93 at 12 m and 1.00 from 15 m (Table IV); these values are optimistic, because our noise is white in time (Section VI).

Wider spacings. The dilation is a fixed fraction of the baseline, whereas the noise floor decays as $1/g$, so wider formations or lower-noise fixes (RTK) favour the two-sided tests. In supplementary runs at 50 and 100 m spacing (20 missions each, same attack model and seeds), the two-sided majority test reaches $P_{\mathrm{D}} = 1.00$, the worst-pair test 0.56 and 0.65 (AUC 0.86 and 0.96), and the collapse detectors remain at zero.

E. Sensitivity to the Attack Geometry

Array enclosure. The attack requires an array that encloses the operating area, and what fails outside it is the azimuth routing, not the elevation mismatch. With a 2.5 km array centred on the start point the swarm leaves the circle for 75 % of the mission. While it is inside, the routing matches the satellite azimuths to within 4.6° (median) and the signed dilation is +24.9 %, the same regime as the enclosing 10 km array (+22 %), as expected from (5), which does not depend on the stand-off distance. Once the swarm is outside, every transmitter lies in a narrow angular sector behind it, the median azimuth mismatch of the routing rises to 83.5°, and

TABLE II

DETECTION PROBABILITY AT $P_{\mathrm{FA}} = 10^{-2}$ PER EPOCH AGAINST THE DISTRIBUTED GROUND ATTACK ($\sigma_{\mathrm{pr}} = 3\,\mathrm{m}$, $\sigma_{\mathrm{rng}} = 0.1\,\mathrm{m}$; 20 MISSIONS, 4800 EPOCHS PER CELL; 95 % CI FROM THE PAIRED MISSION BOOTSTRAP).

g [m]	Jansen min	Jansen max	Jansen majority	Worst-pair	Two-sided majority
5	0.005 [0.004, 0.011]	0.007 [0.004, 0.009]	0.000 [0.000, 0.000]	0.029 [0.016, 0.048]	0.382 [0.215, 0.569]
8	0.002 [0.000, 0.004]	0.003 [0.002, 0.005]	0.000 [0.000, 0.000]	0.065 [0.024, 0.119]	0.572 [0.386, 0.780]
10	0.002 [0.000, 0.004]	0.002 [0.001, 0.005]	0.000 [0.000, 0.000]	0.113 [0.044, 0.195]	0.632 [0.461, 0.823]
12	0.001 [0.000, 0.005]	0.001 [0.000, 0.003]	0.000 [0.000, 0.000]	0.153 [0.064, 0.268]	0.713 [0.577, 0.871]
15	0.002 [0.000, 0.004]	0.001 [0.000, 0.001]	0.000 [0.000, 0.000]	0.193 [0.088, 0.335]	0.845 [0.729, 0.947]
20	0.001 [0.000, 0.003]	0.000 [0.000, 0.001]	0.000 [0.000, 0.000]	0.262 [0.131, 0.426]	0.975 [0.946, 0.997]

TABLE III

ROC AREA (AUC). 0.5 IS CHANCE; VALUES BELOW 0.5 MEAN THAT THE ATTACK MOVES THE STATISTIC AWAY FROM THE ALARM REGION (SEPARABLE, BUT IN THE OTHER TAIL).

g [m]	J. min	J. max	J. maj.	Worst-pair	2-sided maj.
5	0.312	0.430	0.171	0.570	0.829
8	0.262	0.378	0.072	0.622	0.928
10	0.221	0.349	0.037	0.651	0.963
12	0.189	0.324	0.019	0.676	0.981
15	0.156	0.293	0.006	0.706	0.994
20	0.122	0.255	0.001	0.743	0.999

TABLE IV

TWO-SIDED TESTS WITH 1-MIN TEMPORAL INTEGRATION (MOVING MEAN OVER 12 EPOCHS): P_{D} AT $P_{\mathrm{FA}} = 10^{-2}$ PER EPOCH, 95 % CI.

g [m]	Worst-pair, 1-min	Two-sided majority, 1-min
5	0.117 [0.019, 0.205]	0.521 [0.292, 0.701]
8	0.158 [0.055, 0.309]	0.698 [0.491, 0.851]
10	0.182 [0.059, 0.308]	0.804 [0.649, 0.945]
12	0.224 [0.079, 0.382]	0.932 [0.832, 0.991]
15	0.280 [0.102, 0.454]	1.000 [0.994, 1.000]
20	0.376 [0.167, 0.602]	1.000 [1.000, 1.000]

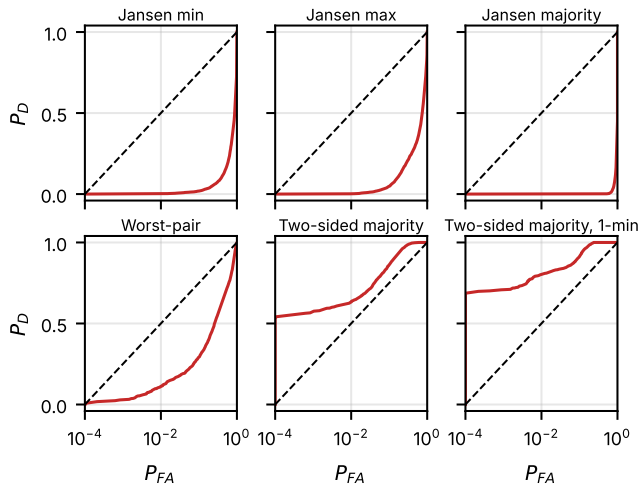


Fig. 7. ROC curves at $g = 10\,\mathrm{m}$ against the distributed ground attack (logarithmic P_{FA} axis; dashed: chance). The collapse detectors lie below the chance line, i.e. separable but in the opposite tail; the two-sided tests lie above it.

the formation is no longer dilated but strongly contracted ($-46\,\%$), which a two-sided test detects immediately. The rest of this subsection uses the enclosing 10 km array and varies the transmitter altitude.

Transmitter altitude. By (5), the dilation decreases as the transmitters are raised towards the satellite elevations, and it vanishes for every satellite only if each transmitter lies on its satellite's line of sight, the placement realized at altitude by GNSS-WASP [11]. We quantify this trade-off at $g = 10\,\mathrm{m}$, the centre of the spacing range, on the same 20 missions (same seeds) as the main results. The dilation D is measured on noiseless runs; the noisy runs serve only to measure P_{D}. Elevated transmitters use a free-space link instead of the ground channel of (3), with the EIRP raised to 10 mW. Capture is complete in every mission (minimum margin 13.9 dB) and, since the delays do not depend on the channel, the ground point of the sweep coincides with the main results.

The 24-transmitter, 10 km array is raised from the ground to 10 km, i.e. from $\theta_k \approx 0^\circ$ to 47° as seen from the swarm (Fig. 8). The first-order model (6) reproduces the measured distortion at every altitude (top panel). The median signed dilation decreases monotonically from $+22\,\%$ on the ground to $+15.5\,\%$ at 19° and $+8.8\,\%$ at 27°, crosses zero at $\theta_k \approx 36^\circ$ (7 km of altitude for this array), and becomes a *contraction* of $-15\,\%$ at 47°. With a single altitude for all transmitters the residual cannot vanish for every satellite at once: satellites below θ_k pull the formation inward, while those above still push it outward. The spread across missions is large; at the crossing the 10th–90th percentiles of D are $-15\,\%$ and $+62\,\%$, depending on the elevations of the satellites in view. For this reason the distortion E pooled over missions is smallest ($10.7\,\%$) before the crossing, at 27°, and is $14.6\,\%$ at the crossing itself.

The detectors follow the sign of the residual (middle panel). The two-sided majority test decreases from $P_{\mathrm{D}} = 0.63$ on the ground to a minimum of 0.39 at the crossing and rises again to 0.59 at 47° (0.80, 0.53 and 0.71 with one-minute integration). The collapse test, blind on the ground, starts firing as the formation contracts ($P_{\mathrm{D}} = 0.22$ at 27°, 0.44 at the crossing, 0.64 at 47°). The worst-pair tests stay at 0.05–0.18 throughout, because the residual that survives at the crossing, 10–15 % of the baseline with a sign that varies across pairs and missions, remains below their noise. The minimum of the aggregate test is not lower because the crossing altitude

TABLE V

LOITERING MISSION ($g = 10$ M, 20 MISSIONS): GROUND CIRCLE VS. STATIC LINE-OF-SIGHT (WASP-LIKE) PLACEMENT AT SLANT RANGE R_{los}. SIGNED DILATION D AND DISTORTION E IN % (NOISELESS); P_{D} AT $P_{\mathrm{FA}} = 10^{-2}$ WITH NOISE (AUC IN PARENTHESES).

Placement	D [%]	E [%]	J. maj.	Worst-pair	2-sided maj.
Ground circle, 10 km	+20.5	20.5	0.00 (0.04)	0.11 (0.65)	0.62 (0.96)
LOS, $R_{\mathrm{los}} = 2$ km	+5.0	11.0	0.13 (0.34)	0.04 (0.56)	0.29 (0.75)
LOS, $R_{\mathrm{los}} = 5$ km	+0.9	5.8	0.10 (0.41)	0.04 (0.54)	0.23 (0.65)
LOS, $R_{\mathrm{los}} = 10$ km	+0.3	4.3	0.09 (0.44)	0.04 (0.53)	0.18 (0.60)

depends on the mission: at any single altitude, most missions are still dilated or already contracted.

The bottom panel makes this explicit by plotting P_{D} of every mission and configuration against its dilation D. The two-sided majority test is blind for $D \in [-8, +5]$ % ($P_{\mathrm{D}} < 0.1$) and detects outside this window (0.71 for D between +12 and +20 %, 0.93 beyond +20 %, 0.81 beyond −20 %); the collapse test is its mirror image on the contraction side. The window is shifted towards contraction by the noise bias of the median ratio. Under H_0 the median ratio is $1 + b_0$, with $b_0 \approx 0.045$ at 10 m because noise lengthens distances on average; under attack it is approximately $1 + D + b_0$. The attacked statistic $|D + b_0|$ thus falls within the H_0 spread for D slightly negative, and a contraction of a few percent reads as no attack. Centring the test on the calibrated H_0 median $1 + b_0$ instead of one is expected to re-centre the window on $D = 0$. With a common-altitude array, the attacker's optimum is therefore a narrow, mission-dependent band around the zero crossing (about 7 km of altitude for a 10 km array), not the highest altitude. Reaching it in every mission would require per-satellite altitudes, i.e. the line-of-sight placement.

Line-of-sight placement. The second experiment is a *static* version of the GNSS-WASP geometry: one transmitter per satellite visible at the start of the mission, placed on its line of sight from the array centre at slant range R_{los} (altitude $R_{\mathrm{los}} \sin \mathrm{el}_s$, up to 9 km for $R_{\mathrm{los}} = 10$ km). The constellation is built once, from the satellites visible at the first epoch; a satellite that rises later is routed by (2) to the transmitter closest in azimuth, which sits on another satellite's line of sight, so it is spoofed but with a larger residual. Such a constellation is on the lines of sight only for a swarm near the reference point, as in GNSS-WASP, whose bias grows with the distance from it. The experiment therefore uses a loitering mission confined within 700 m of the centre, and compares it with the ground circle on the same mission (Table V). The line-of-sight placement cancels the signed dilation (+20.5 % for the ground circle; +5.0, +0.9 and +0.3 % at $R_{\mathrm{los}} = 2$, 5 and 10 km) but leaves a residual distortion of 11, 5.8 and 4.3 % of varying sign, because the transmitters are fixed while the satellites move. The two-sided majority test drops from $P_{\mathrm{D}} = 0.62$ (AUC 0.96) to 0.29, 0.23 and 0.18 (AUC 0.75, 0.65 and 0.60), the worst-pair test to 0.04, and the collapse test fires in about a tenth of the epochs on the contracting missions. GNSS-WASP's transmitters track the satellites, so this static residual is an upper bound on the WASP error, and its detectability an upper bound on WASP's. A constellation on the lines of sight at slant ranges of 5 km and beyond is detected in fewer than a quarter of the epochs by the best relative-distance test at this spacing, against about two thirds for the ground array, at the price of airborne platforms several kilometres high.

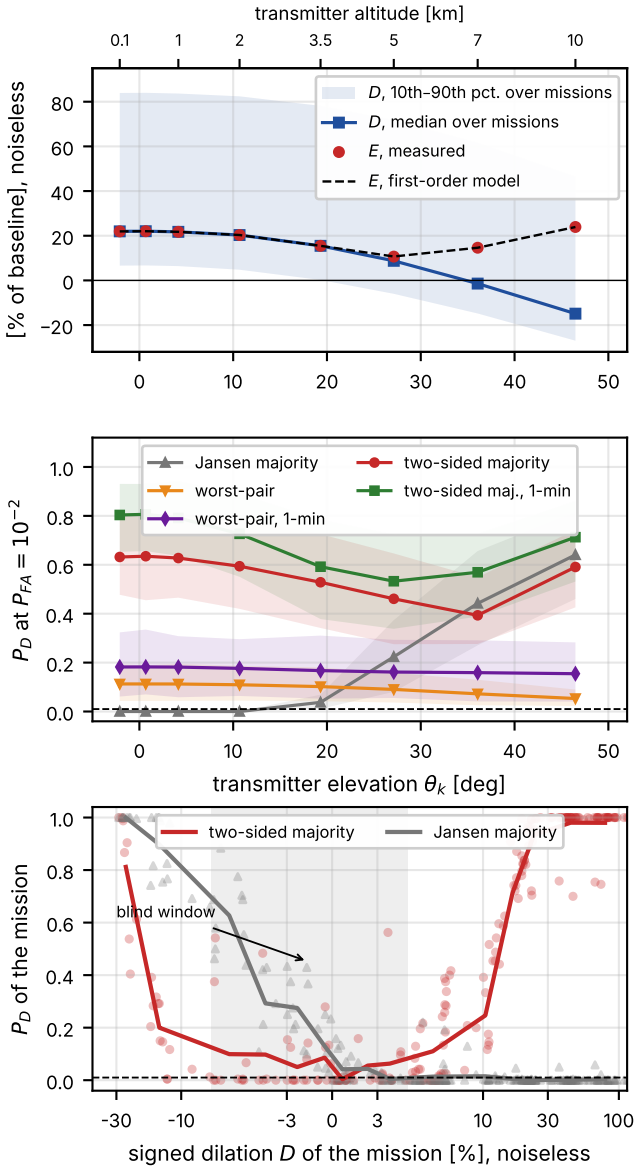


Fig. 8. Transmitter altitude sweep (24 transmitters, 10 km array, $g = 10$ m, 20 missions). Top: signed dilation D (median and 10th–90th percentiles over the missions) and distortion E (measured, and predicted by the first-order model (6)), all from noiseless runs. Middle: P_{D} at $P_{\mathrm{FA}} = 10^{-2}$ with noise (bands: 95 % CI). Bottom: P_{D} of each mission against its dilation D, for all altitudes and placements; the two-sided majority test is blind for $D \in [-8, +5]$ %, and the collapse test fires only for contractions.

VI. Discussion and Limitations

What a ground array can and cannot do: A distributed ground spoofer defeats the premise of the classical multi-receiver countermeasure, since it does not collapse the formation, but it cannot preserve the formation either: the elevation mismatch of (5) dilates it by a median of 22 % of the baseline, independently of the transmitter power and of the formation spacing. For the defender this is a change of signature, not a loss of observability. Collapse tests should be replaced, or at least complemented, by two-sided aggregate tests on the distance ratios. In our simulation these detect the dilation reliably at 15–20 m and in a majority of epochs at 8–12 m with standard positioning, and reliably from 12–15 m with one minute of integration. Raising the transmitters reduces and then reverses the dilation, so a defender should test the distance ratios on *both* sides and expect a blind window of a few percent of dilation around zero. A spoofer that compensates the residual with per-satellite altitudes is left to future work.

Noise model and operating point: Our pseudorange noise is independent across receivers and white in time. In practice, a large part of the GNSS error (ionosphere, troposphere, ephemeris) is common to receivers a few metres apart, with correlation coefficients of 0.4–0.6 measured in [7], and it is correlated over minutes. Spatial correlation lowers the noise floor of every relative-distance test, including the worst-pair test, so our worst-pair results are pessimistic. Temporal correlation reduces the gain of a moving average, so our one-minute integration results (Table IV) are optimistic. The per-epoch false-alarm rate of 10^{-2} corresponds to about two false alarms per 20-minute mission; an operational detector would need a mission-level false-alarm constraint, e.g. with a sequential test. Finally, the UWB ranges are modelled as unbiased. A calibration or non-line-of-sight bias of a few decimetres is a few percent of a 5–10 m baseline, of the same order as the blind window of the two-sided majority test, and would have to be calibrated out.

Scope of the result: The evaluation covers detectors that are functions of relative positions or distances, over the spacings of GNSS-based formation flight. It does *not* extend to (i) *direction-of-arrival* detectors: a circle of ground transmitters produces arrivals at near-zero elevation, inconsistent with the satellite constellation, so multi-antenna arrays with attitude information [3], [17] would detect our attacker unless its transmitters are placed along the satellite directions as in [11]; (ii) *absolute-reference* checks, such as ranging to fixed infrastructure, TDoA from ground sensor networks [14], or terrain and signal-of-opportunity positioning, which directly observe a displacement of several kilometres; (iii) GNSS/INS detectors such as SemperFi [18] and INS innovation monitors [19], [20], which respond to the time variation of the induced error and were not evaluated.

Relation to formation-preserving attacks: Tippenhauer et al. [6] prove that a formation-preserving placement exists, GNSS-WASP [11] approximates it with transmitters in line with the satellites at altitude, and Park and Yoo [25] show that an exactly common translation is invisible to any relative-only detector. Our attacker is more constrained, and the dilation we measure and model is what separates it from those ideal attacks. Section V-E connects the two regimes: raising a common-altitude array changes the sign of the residual rather than removing it, and only a constellation on the lines of sight at several kilometres of slant range approaches the regime of [11]. Even then, a static constellation leaves a 4–6 % residual over a 20-minute mission, because the satellites move while the transmitters do not.

Other simplifications: The result assumes an array that encloses the swarm's operating area; a static array that the swarm leaves deforms the formation by more than 50 % and is readily detected. The decoy position is fixed and the attack is modelled in its steady state after capture; the capture transient, time-varying decoys and the routing switches of (2) are not modelled. The UAVs fly open loop: in a real swarm, guidance would follow the spoofed position, and a GNSS-based formation controller would react to the computed dilation by contracting the true formation. The attacker is assumed to know the true position of the swarm centre at every epoch. The simulator uses a spherical Earth, a synthetic single-constellation geometry with a 0° elevation mask, no multipath or shadowing, and signal capture by power comparison without tracking-loop dynamics. We did not vary the swarm size m or the number of transmitters N_{sp}; with few UAVs, the median over pairs averages less noise and the two-sided majority test loses power. Twenty missions per spacing bound P_{D} to within about ± 0.1 for the two-sided tests; the conclusions on the collapse detectors do not depend on this resolution.

VII. Conclusion

We simulated a distributed ground spoofer, with 24 fixed transmitters, one impersonated satellite each, three-dimensional propagation and a ground-to-UAV channel, attacking a UAV swarm at the 5–20 m spacings of outdoor GNSS-based formation flight, and evaluated relative-distance detectors against it. The attack displaces the swarm by about 4 km but cannot preserve its formation: the mismatch between satellite and transmitter elevations dilates the computed formation by a median of 22 % of the baseline, which a first-order least-squares model reproduces epoch by epoch. The dilation lies in the tail that the collapse detectors of Jansen et al. do not test ($P_{\mathrm{D}} \leq 0.011$, AUC down to 0.001), and noise largely masks it for the worst-pair GNSS-vs-UWB check ($P_{\mathrm{D}} \leq 0.26$). A two-sided test on the median distance ratio detects it in a majority of epochs from 8 m and reliably from 15 m (P_{D} from 0.38 at 5 m to 0.98 at 20 m). Raising the transmitters does not remove the residual but reverses it. The attacker's best common altitude is a narrow, mission-dependent band around the zero crossing (about 7 km for a 10 km array), and only a constellation on the satellites' lines of sight at several kilometres of slant range comes close to undetectability. Relative-distance detection against distributed ground spoofers should therefore test the distance ratios for

deviations on both sides rather than for collapse, and should be complemented by direction-of-arrival or absolute-reference information against airborne attackers that can also reproduce the elevations.

ACKNOWLEDGMENTS

[TODO: ringraziamenti, funding]

REFERENCES

[1] European Union Agency for the Space Programme (EUSPA), "Galileo open service navigation message authentication (OSNMA): Initial service declaration," https://www.euspa.europa.eu/galileo-osnma, 2025, initial service declared operational on 24 July 2025.

[2] T. E. Humphreys, B. M. Ledvina, M. L. Psiaki, B. W. O'Hanlon, and P. M. Kintner, Jr., "Assessing the spoofing threat: Development of a portable GPS civilian spoofer," in *Proceedings of the 21st International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS 2008)*, Savannah, GA, USA, 2008, pp. 2314–2325.

[3] M. L. Psiaki and T. E. Humphreys, "GNSS spoofing and detection," *Proceedings of the IEEE*, vol. 104, no. 6, pp. 1258–1270, 2016.

[4] A. J. Kerns, D. P. Shepard, J. A. Bhatti, and T. E. Humphreys, "Unmanned aircraft capture and control via GPS spoofing," *Journal of Field Robotics*, vol. 31, no. 4, pp. 617–636, 2014.

[5] A. Altaweel, H. Mukkath, and I. Kamel, "GPS spoofing attacks in FANETs: A systematic literature review," *IEEE Access*, vol. 11, pp. 55 233–55 280, 2023.

[6] N. O. Tippenhauer, C. Pöpper, K. B. Rasmussen, and S. Čapkun, "On the requirements for successful GPS spoofing attacks," in *Proceedings of the 18th ACM Conference on Computer and Communications Security (CCS)*. ACM, 2011, pp. 75–86.

[7] K. Jansen, N. O. Tippenhauer, and C. Pöpper, "Multi-receiver GPS spoofing detection: Error models and realization," in *Proceedings of the 32nd Annual Conference on Computer Security Applications (ACSAC)*. ACM, 2016, pp. 237–250.

[8] P. F. Swaszek and R. J. Hartnett, "Spoof detection using multiple COTS receivers in safety critical applications," in *Proceedings of the 26th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2013)*, Nashville, TN, USA, 2013, pp. 2921–2930.

[9] P. Mykytyn, M. Brzozowski, Z. Dyka, and P. Langendoerfer, "GPS-spoofing attack detection mechanism for UAV swarms," in *2023 12th Mediterranean Conference on Embedded Computing (MECO)*, 2023.

[10] S. Bi, K. Li, S. Hu, W. Ni, C. Wang, and X. Wang, "Detection and mitigation of position spoofing attacks on cooperative UAV swarm formations," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1883–1895, 2024.

[11] C. Tibaldo, H. Sathaye, G. Camurati, and S. Čapkun, "GNSS-WASP: GNSS wide area SPoofing," in *Proceedings of the 34th USENIX Security Symposium*. USENIX Association, 2025, pp. 7039–7058.

[12] A. Al-Hourani and K. Gomez, "Modeling cellular-to-UAV path-loss for suburban environments," *IEEE Wireless Communications Letters*, vol. 7, no. 1, pp. 82–85, 2018.

[13] P. F. Swaszek, R. J. Hartnett, and K. C. Seals, "GNSS spoof detection using independent range information," in *Proceedings of the 30th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2017)*, Portland, OR, USA, 2017, pp. 2838–2853.

[14] K. Jansen, M. Schäfer, D. Moser, V. Lenders, C. Pöpper, and J. Schmitt, "Crowd-GPS-Sec: Leveraging crowdsourcing to detect and localize GPS spoofing attacks," in *2018 IEEE Symposium on Security and Privacy (SP)*, 2018, pp. 1018–1031.

[15] N. Stenberg, E. Axell, J. Rantakokko, and G. Hendeby, "Results on GNSS spoofing mitigation using multiple receivers," *NAVIGATION: Journal of the Institute of Navigation*, vol. 69, no. 1, p. navi.510, 2022.

[16] A. Kalantari and E. G. Larsson, "Statistical test for GNSS spoofing attack detection by using multiple receivers on a rigid body," *EURASIP Journal on Advances in Signal Processing*, vol. 2020, p. 8, 2020.

[17] D. Borio and C. Gioia, "A sum-of-squares approach to GNSS spoofing detection," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 52, no. 4, pp. 1756–1768, 2016.

[18] H. Sathaye, G. LaMountain, P. Closas, and A. Ranganathan, "SemperFi: Anti-spoofing GPS receiver for UAVs," in *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2022.

[19] C. Tanil, S. Khanafseh, M. Joerger, and B. Pervan, "An INS monitor to detect GNSS spoofers capable of tracking vehicle position," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 54, no. 1, pp. 131–143, 2018.

[20] B. Kujur, S. Khanafseh, and B. Pervan, "Detecting GNSS spoofing using temporal behavior of innovation sequences," *NAVIGATION: Journal of the Institute of Navigation*, vol. 71, no. 1, p. navi.629, 2024.

[21] W. Chen, Y. Dong, and Z. Duan, "Impacts of a single GPS spoofer on multiple receivers: Formal analysis and experimental evaluation," in *2024 IEEE 21st Consumer Communications & Networking Conference (CCNC)*, 2024, pp. 127–134.

[22] ——, "GPS swarm spoofing: Theoretical analysis and practical solutions," in *Proceedings of the IEEE International Conference on Communications (ICC)*, 2025, pp. 3345–3350.

[23] M. Ceccato, F. Formaggio, and S. Tomasin, "Spatial GNSS spoofing against drone swarms with multiple antennas and Wiener filter," *IEEE Transactions on Signal Processing*, vol. 68, pp. 5782–5794, 2020.

[24] X. Cheng, H. Yang, S. Liu, and Y. Yang, "Distributed multi-antenna GPS spoofing attack using off-the-shelf devices," in *Proceedings of the 18th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*. ACM, 2025, pp. 28–39.

[25] M. Park and J. S. Yoo, "Rigid-covert GNSS spoofing of UAV swarms: A structural blind spot, its detection limit, and absolute-anchor defenses," arXiv:2608.06885, 2026, preprint.

[26] J. Aspnes, T. Eren, D. K. Goldenberg, A. S. Morse, W. Whiteley, Y. R. Yang, B. D. O. Anderson, and P. N. Belhumeur, "A theory of network localization," *IEEE Transactions on Mobile Computing*, vol. 5, no. 12, pp. 1663–1678, 2006.

[27] A. Sebastian and B. C. Manjith, "A comprehensive survey of GPS spoofing attack detection techniques in UAV systems," in *2026 IEEE International Conference on Intelligent Technologies, Systems and Innovation Frontiers (ICITSIF)*, 2026, pp. 1041–1046.

[28] S. Schilleci *et al.*, "[title — complete from ieee xplore document 11587831]," in *Proc. IEEE Conf.*, 2025, iEEE Xplore document no. 11587831; complete citation fields before submission.

[29] S. M. Kay, *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory*. Upper Saddle River, NJ, USA: Prentice Hall, 1998.

[30] J. H. Cheok, V. R. Aparow, J. Z. N. Ng, J. L. Cheah, and D. Leong, "Development and optimization of formation flying for unmanned aerial vehicles using particle swarm optimization based on reciprocal velocity obstacles," *SAE International Journal of Aerospace*, vol. 15, no. 2, pp. 171–184, 2022.

[31] P. Guo, R. Zhang, and B. Xu, "Safety separation distance design for UAV formation based on system performance," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 3, pp. 5980–5995, 2025.